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Statistical Modelling and Inference

Week 5, Lecture 2

Outline:

- i. Multiple Linear Regression
- ii. Least Square Estimation
- iii. ANOVA



Review of Simple Linear Regression Model

- Only **one independent variable**, X
- Relationship between X and Y is described by a linear function
- Changes in Y are assumed to be caused by changes in X

$$Y = \beta_0 + \beta_1 X + \varepsilon$$



Multiple Linear Regression Model

- **More than one** independent variables, X_i
- Relationship between X_i and Y is described by a linear function
- Changes in Y are assumed to be caused by changes in X_i

$$Y = \beta_0 + \beta_i X_i + \varepsilon$$

or

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$



Least Squares Criterion

- β_0 and β_1 are obtained by finding the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of the squared residuals.

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$
$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$



The Least Squares Estimates

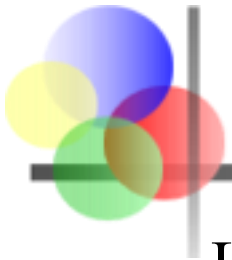
The formulas for Slope is $\hat{\beta}_1$ and intercept $\hat{\beta}_0$ are:

$$\hat{\beta}_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

or,

$$\hat{\beta}_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$



The Least Squares Estimates (Matrix approach)

Let's consider a model having k independent variables.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

In matrix form can be written as:

$$Y = X\beta + \varepsilon$$

Where,

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}_{n \times (k+1)} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix}_{(k+1) \times 1} \quad \varepsilon = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_n \end{bmatrix}_{n \times 1}$$



The Least Squares Estimates (Matrix approach)

Estimation of vector $\hat{\beta}$:

$$Y = X\beta$$

$$X^t Y = X^t X \beta$$

$$(X^t X)^{-1} X^t Y = (X^t X)^{-1} X^t X \beta$$

$$\hat{\beta} = (X^t X)^{-1} X^t Y$$

Also, you can estimate the values of dependent variable by using:

$$\hat{Y} = X \hat{\beta}$$



Example:

The price (y) in \$1000AUD, age (x_1) in decades, and area (x_2) in 1000m², of five randomly selected houses in a regional South Australian town are shown in the table below.

y	100	80	104	94	130
x_1	1	5	5	10	20
x_2	1	1	2	2	3

- a) Fit a multiple linear regression model to this data set.
- b) Test the hypothesis $H_0: \beta_1 = \beta_2 = 0$ against at 5% level of significance.



The Least Squares Estimates (Matrix approach)

a) Let's consider a model having two independent variables.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$$

In matrix form can be written as:

$$Y = X\beta + \varepsilon$$

Where,

$$Y = \begin{bmatrix} 100 \\ 80 \\ 104 \\ 94 \\ 130 \end{bmatrix}$$

and,

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 5 & 2 \\ 1 & 10 & 2 \\ 1 & 20 & 3 \end{bmatrix}$$



The Least Squares Estimates (Matrix approach)

$$X^t X = \begin{bmatrix} 5 & 41 & 9 \\ 41 & 551 & 96 \\ 9 & 96 & 19 \end{bmatrix} \quad X^t Y = \begin{bmatrix} 508 \\ 4560 \\ 966 \end{bmatrix}$$

$$(X^t X)^{-1} = \frac{1}{543} \begin{bmatrix} 1253 & 85 & -1023 \\ 85 & 14 & -111 \\ -1023 & -111 & 1074 \end{bmatrix}$$

Hence the least square estimates matrix is:

$$\hat{\beta} = (X^t X)^{-1} X^t Y = \begin{bmatrix} 66.1252 \\ -0.3794 \\ 21.4365 \end{bmatrix}$$



The Least Squares Estimates (Matrix approach)

Hence the least square estimates are:

$$\hat{\beta}_0 = 66.1252$$

$$\hat{\beta}_1 = -0.3794$$

$$\hat{\beta}_2 = 21.4365$$

Finally the estimated regression model is:

$$\hat{y} = 66.1252 - 0.3794x_1 + 21.4365x_2$$



ANOVA Summary

b) Test the hypothesis $H_0: \beta_1 = \beta_2 = 0$ at 5% level of significance.

$$H_0: \beta_1 = \beta_2 = 0 \quad \text{and} \quad H_a: \beta_1 \neq \beta_2 \neq 0$$

Source	df	SS	MSS	F	P-value
x_1	1	724.17	724.17	3.7845	0.1911
x_2	1	232.33	232.33	1.2141	0.3854
Error	2	382.70	191.35		
Total	4				

Conclusion: Because both p-values are greater than level of significance: 0.05, so we can not reject the null hypothesis $H_0: \beta_1 = \beta_2 = 0$.



**Thank
You**

谢谢

